

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q14-0329

Omni-Hackberry 500 kV
May 03, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0329. The Interconnection Customer proposes a 869.7 MW natural gas combined cycle facility interconnecting at the Omni-Hackberry 500 kV substation in the Ohio area. The request was received on April 27, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 155 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$7,796,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
W Canton 500/230kV TX	Point Arena to Weedpatch 500.0 kV L	114.9	1,168	8.7
Old Forge to Portage Lakes 230.0 kV line	Surrowiec to Weedpatch 500.0 kV	107.5	833	7.7
Gavin Power Llc to Olympus 230.0 kV	Gavin Power Llc 500 kV/345 kV XFMR	102.1	611	8.2
North Cambridge to W Canton 230.0 kV	Weedptch to Point Area 500.0 kV L	101.2	775	6.4

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
W Canton 500/230kV TX	Replace transformer with higher-rated unit	\$26,124,000	21% (\$5,594,000)
Old Forge to Portage Lakes 230.0 kV line	Reconductor line	\$6,261,000	20% (\$1,268,000)
Gavin Power Llc to Olympus 230.0 kV	Reconductor line	\$3,507,000	10% (\$338,000)
W Canton to N Cambridge 230.0 kV L	Reconductor line	\$2,465,000	24% (\$596,000)
	Total allocated to project		\$7,796,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.